

MODERN SEMANTICS STUDIES IN TRACING THE SHIFT OF LANGUAGE MEANING

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Abstract

This study explores the field of modern semantics with a particular focus on how shifts in language meaning can be traced and understood through a literature-based approach. Language is a dynamic and evolving system, and meaning is not static; it constantly adapts to cultural, social, and technological developments. Modern semantic theories offer various frameworks for analyzing these changes, including componential analysis, prototype theory, and cognitive semantics. Through the lens of these frameworks, this study examines how meaning shifts occur over time, such as through broadening, narrowing, amelioration, pejoration, and semantic bleaching. The literature reveals that language change often reflects deeper transformations within society, where words take on new meanings or lose their original significance. Modern semantics provides the analytical tools to decode these transformations systematically. The study also emphasizes the role of context, usage patterns, and pragmatic influence in shaping meaning. Advances in linguistics have allowed for a more nuanced understanding of lexical change, especially through corpus-based studies and discourse analysis. By relying solely on library research, this study synthesizes existing knowledge to identify patterns and trends in meaning change across various languages and contexts. It concludes that modern semantics is a vital discipline for tracing how language evolves and adapts. Understanding these shifts enhances not only our comprehension of language itself but also offers insights into the broader human experience reflected in linguistic expression. This research underlines the importance of continued theoretical exploration in semantics to keep pace with the dynamic nature of language.

Keywords: Modern Semantics, Language Change, Meaning Shift, Lexical Semantics, Semantic Analysis.

INTRODUCTION

Language is a living, evolving construct that reflects and refracts the social, cultural, and intellectual environments in which it is used. At the heart of linguistic evolution lies semantics the study of meaning. In modern linguistic inquiry, semantic change has emerged as a central theme, reflecting how the lexicon adapts to human experience, technological advancement, and ideological shifts. As language users engage with an increasingly complex world, they

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naturally extend, narrow, invert, or otherwise alter meanings, often unconsciously. Modern semantic theories, shaped by cognitive science, formal logic, pragmatics, and computational modeling, attempt to explain the mechanisms behind these changes and provide frameworks for tracing them. These frameworks allow researchers to discern patterns and tendencies in meaning shifts, revealing how language maps onto changing human realities (Feist, 2022).

Semantic change, or the evolution of word meanings over time, is not a new phenomenon; however, the tools and theories for analyzing such changes have undergone substantial transformation. In the past, traditional approaches such as historical linguistics and etymology served as the primary means of tracing lexical evolution. Today, modern semantics incorporates distributional semantics, corpus linguistics, psycholinguistics, and computational modeling to detect subtle and large-scale semantic shifts. This interdisciplinary approach has opened new vistas in understanding how language responds to socio-political movements, technological disruptions, and cognitive constraints (Geeraerts et al., 2024). These advances underscore the value of semantics not only as a branch of linguistics but as a powerful interpretative tool for understanding human cognition and cultural change.

Modern semantics does not exist in isolation; it is shaped by broader trends in linguistics and philosophy. The distinction between sense and reference, the role of context in interpretation, and the interplay between lexical semantics and syntax all form part of the rich tapestry within which semantic analysis operates. For instance, prototype theory has contributed to understanding how meaning categories are formed not through rigid definitions but by family resemblances and typicality effects. This has enabled researchers to better analyze semantic shifts that arise when core prototypes evolve or when peripherally related concepts become central (Reilly et al., 2025). Such theories have particular relevance in analyzing contemporary semantic phenomena such as metaphorical extension, jargon formation, and semantic bleaching.

Recent research highlights the increasing role of computational tools in tracing meaning shifts. Embedding models such as BERT, GPT, and word2vec can represent the semantics of words in high-dimensional vector spaces, allowing for precise measurement of how meanings evolve over time through contextual changes. This has been particularly impactful in analyzing social media data, where rapid cultural shifts are reflected almost instantaneously in language. For example, during the COVID-19 pandemic, numerous studies used word embeddings to track how words like "isolation," "mask," and "positive" shifted semantically in digital discourse (Guo et al., 2021). These models offer empirical grounding to the theory of semantic drift,

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providing evidence that language change is not just conceptual but detectable through measurable shifts in usage patterns.

The phenomenon of semantic shift can be categorized in various ways—broadening, narrowing, amelioration, pejoration, metaphorical extension, and more. Each of these reflects different cognitive and social processes. Broadening, for instance, occurs when a word extends its scope to encompass additional meanings, such as how "mouse" came to refer not only to a rodent but also a computer accessory. Narrowing is its converse, seen in the historical development of "meat," which once referred to all food but now denotes animal flesh. These processes are often the result of metaphor, metonymy, synecdoche, or euphemism, and they demonstrate how users continually adapt language to meet communicative needs (Koslow, 2022). Understanding these mechanisms is crucial for linguists aiming to trace meaning change within historical corpora or contemporary datasets.

Moreover, socio-political contexts play an important role in driving semantic shifts. Language is inherently ideological and reflective of power structures; hence, word meanings often change to reflect dominant ideologies. The evolution of terms such as "gender," "liberty," or "equity" in political discourse illustrates how lexical meanings can shift drastically over short periods in response to shifting societal norms. Such transformations are not merely lexical but represent deeper semiotic and cultural reconfigurations (Aronowitz, 2023). In this light, modern semantic study becomes not just a linguistic exercise but a window into the shifting values of society.

The relationship between semantic change and multilingualism is also central to modern semantics. With globalization and the proliferation of digital communication, languages are increasingly in contact. This results in lexical borrowing, code-switching, and translanguaging, all of which contribute to semantic evolution. For example, in regions like Zambia, public signage and advertising often reflect hybrid linguistic forms that influence meaning perception and transformation (Costley et al., 2023). This interaction across languages produces novel semantic fields and challenges traditional notions of fixed meaning. Modern semantics thus engages with language as a fluid, porous system that constantly negotiates meaning in multilingual contexts.

Modern semantics also benefits from historical insights. The Middle English period, for example, was a fertile ground for lexical borrowing and meaning shift, as shown by the absorption of French and Latin terms into English. Semantic analysis of this period reveals not only lexical expansion but also shifts in conceptual categorization. Words such as "court," "virtue," or "justice" entered English with nuanced meanings, often reflecting the prestige and

administrative structures of Norman society (Sylvester et al., 2022). Understanding these historical shifts provides a foundation for analyzing contemporary changes and highlights the cyclical nature of meaning evolution.

One of the key challenges in modern semantics is distinguishing between stable meanings and those undergoing transformation. Semantic stability refers to the persistence of a core meaning across time and context, while semantic modulation involves shifts due to discourse context, collocation, or pragmatic inference. Researchers have increasingly turned to computational tools to analyze these dynamics. BERT embeddings, for instance, allow scholars to detect contextualized meaning variations, revealing how the same word may carry different connotations in different usage environments (Cassani et al., 2023). This challenges the classical assumption of discrete word senses and supports a more fluid, dynamic view of meaning.

Another area of growing importance is the role of technology in shaping and being shaped by semantic change. In computer vision and artificial intelligence, the semantics of labels used in machine learning models have significant consequences for system reliability. Errors in semantic segmentation, for instance, can lead to misclassifications with real-world impacts (De Jorge et al., 2023). At a broader level, the integration of semantic knowledge into AI systems depends on accurate modeling of how meanings change, not only across contexts but also over time. This highlights the necessity of aligning computational models with linguistic theories of semantics.

Pedagogically, semantic study is also evolving. Modern educational environments are now incorporating dynamic tools to teach abstract semantic principles. Interactive simulations, conceptual modeling platforms, and real-time corpus analysis tools are used to engage students with real-world data and semantic patterns. These methods allow for a more intuitive understanding of how language meaning shifts in various domains, from literature and media to scientific discourse (Steingartner & Sivý, 2024). The intersection of education, cognition, and semantics is opening new pathways for understanding how meaning is acquired, transformed, and applied.

The use of semantic change detection tools, particularly in under-resourced languages, has further broadened the scope of modern semantic inquiry. Projects such as RuShiftEval focus on semantic drift in Russian, combining distributional semantics with historical corpora to trace nuanced changes in lexical usage (Kutuzov & Pivovarova, 2021). This approach is valuable not only for linguistic analysis but also for preserving cultural heritage. As language changes,

it also risks erasing older conceptualizations; thus, semantic preservation becomes a form of cultural preservation. By identifying and documenting semantic shifts, researchers contribute to a more comprehensive understanding of linguistic and cultural continuity.

It is also crucial to acknowledge the philosophical underpinnings of semantic study. The nature of meaning, the reference of terms, and the boundary between semantics and pragmatics are all questions that continue to be debated. Some theorists argue for a more usage-based, emergent view of meaning, while others maintain a formalist stance grounded in logic and structure. Modern semantic research often bridges these divides by employing hybrid models that integrate formal semantics with cognitive and corpus-based insights (Wedgwood, 2021). This pluralistic approach enriches the field and reflects the complexity of human language and thought.

The concept of semantic vulnerability has also emerged in discussions of artificial intelligence and natural language processing. When large language models are trained on biased or unrepresentative data, their semantic interpretations can become skewed or even dangerous. This has led to concerns about the ethical implications of deploying such models in public-facing applications, particularly in sensitive domains like healthcare, law, or politics (McIntosh et al., 2024). These concerns underscore the importance of grounding semantic analysis in robust, ethically sound methodologies that reflect both linguistic diversity and social responsibility.

Semantic shift is not limited to content words; functional elements like prepositions, conjunctions, and discourse markers also undergo change. These shifts, though often subtle, have profound effects on sentence structure and interpretation. For example, the evolving usage of modal verbs in English reflects changes in how speakers express obligation, possibility, and epistemic stance. Such shifts reveal deeper changes in speaker attitudes, cultural norms, and interactional expectations (Mishra, 2023). A comprehensive understanding of semantic change must therefore encompass both lexical and grammatical elements.

Finally, the study of semantic shift through corpus-based approaches provides empirical validation to theoretical claims. Large corpora such as Google Books, COCA, and social media datasets offer a wealth of diachronic and synchronic data. These resources enable researchers to track frequency changes, collocational patterns, and semantic proximity with statistical rigor. Recent surveys have shown that integrating these approaches leads to more accurate and context-sensitive models of semantic change, further supporting the theoretical frameworks proposed by cognitive and formal linguists (Montanelli & Periti, 2023).

Literature Review

The study of semantics in the context of language change has evolved significantly in recent decades, with numerous researchers contributing diverse perspectives, methodologies, and theoretical frameworks. A key focus in recent literature is the detection and analysis of semantic shift—how word meanings evolve due to sociocultural, psychological, and contextual influences. This literature review synthesizes contemporary scholarship on semantic shift, particularly as it is examined through the lenses of lexical theory, cognitive semantics, corpus linguistics, computational modeling, and sociolinguistics.

The foundational understanding of semantic change rests on the classification of types of shifts such as broadening, narrowing, amelioration, pejoration, metaphorical extension, and bleaching. Mishra (2023) provides a thorough account of how such changes have shaped the English lexicon, identifying processes like borrowing and neologism as primary drivers of semantic innovation. Through detailed analysis of diachronic corpora, Mishra argues that the meanings of many English words have undergone transformations not only because of internal linguistic forces but also through cultural diffusion and technological adaptation. Mishra's work reaffirms the idea that meaning is not fixed; it is perpetually influenced by shifting communicative needs and lexical economy.

On a more theoretical level, Feist (2022) develops a holistic semantic framework in *Significance in Language*, in which meaning is not a static assignment to lexical items but a dynamic negotiation shaped by interpretation, context, and cognition. Feist emphasizes that semantic structures are not purely linguistic but cognitive maps that reflect how humans perceive and categorize reality. His work echoes elements of cognitive semantics, particularly prototype theory, suggesting that semantic shift often arises when core category members change due to conceptual realignment. This allows us to interpret shifts not merely as lexical anomalies but as reflections of broader shifts in human cognition and cultural awareness.

Wedgwood (2021) contributes to the semantic discourse by shifting analytical focus from fixed reference to dynamic interactional meaning. In his volume *Shifting the Focus*, Wedgwood contends that traditional semantics often ignores how meaning emerges in use. He explores how conversation, inference, and pragmatic principles play a central role in shaping meaning, particularly in real-time language usage. Semantic shift, in this light, becomes an interactional phenomenon, produced and reinforced by repeated communicative acts rather than isolated

lexical change. His analysis underscores the necessity of bridging semantics with pragmatics and discourse theory.

A rapidly growing body of work is focused on computational detection of semantic shift, facilitated by large-scale data and machine learning models. Montanelli and Periti (2023) provide a comprehensive survey on contextualized semantic shift detection, reviewing state-of-the-art techniques in distributional semantics. They examine how word embedding models like BERT and RoBERTa can capture nuanced meaning variations across time and contexts by observing the movement of word vectors in high-dimensional spaces. Their study reveals that semantic shift is not always gradual or linear—it can be abrupt, domain-specific, or even transient. This granular detection capability represents a significant advancement in semantic research, offering empirical rigor previously unattainable through traditional methods.

In a similar computational approach, Guo, Xypolopoulos, and Vazirgiannis (2021) investigate the impact of global crises on language, focusing on how COVID-19 reshaped semantic fields in real-time through Twitter discourse. Their analysis illustrates how words like "lockdown," "positive," and "quarantine" took on new meanings or emphases within months. By comparing embeddings across different time slices, they detected sharp semantic shifts directly attributable to sociocultural context. This study exemplifies how language not only reflects reality but actively participates in shaping public perception and collective memory during crises.

The work of Geeraerts et al. (2024) represents a broader, theoretically grounded inquiry into lexical variation and semantic change using a distributional semantic approach. By combining statistical models with theoretical insights from lexical semantics, the authors demonstrate how variation in word use reflects deeper, systemic shifts in semantic structure. Their findings suggest that not all lexical changes are driven by novelty or innovation—some are the result of subtle frequency-based reconfiguration of associations within a lexical field. This aligns with cognitive linguistic perspectives, where meaning is seen as emergent from usage patterns rather than determined solely by dictionaries or formal logic.

In support of this view, Reilly et al. (2025) propose a multidisciplinary glossary of semantic concepts, highlighting how different fields conceptualize "meaning." Their work stresses the importance of terminological precision in semantic studies and reveals that even the term "semantic shift" carries different implications across disciplines. The glossary is particularly useful for researchers navigating the intersections of psycholinguistics, computational linguistics, and philosophy of language. It highlights the epistemological complexities of

defining and tracing meaning and provides a much-needed terminological anchor in an otherwise conceptually fluid field.

The pedagogical implications of semantic theory are explored by Steingartner and Sivý (2024), who present innovative methods for teaching semantic concepts in higher education. They advocate for dynamic, environment-based approaches that involve simulation, visual modeling, and context-driven analysis. According to their study, students grasp semantic shift more effectively when exposed to live corpora and real-time lexical data, as opposed to static textbook examples. This suggests that semantics education must evolve alongside advances in computational and corpus-based research, integrating theory with practice in classroom environments.

Historical semantic change remains a critical area of inquiry, especially for understanding long-term patterns in lexical development. Sylvester, Tiddeman, and Ingham (2022) analyze Middle English lexical borrowing and its semantic consequences, revealing how cultural contact introduces not only new words but new conceptual frameworks. They show that semantic outcomes of borrowing are domain-dependent—religious terms tend to retain original meanings, while administrative and legal vocabulary undergo extensive contextual reshaping. Their multi-domain analysis demonstrates that language borrowing is rarely semantically neutral; instead, it often introduces ideological and cultural frames that influence interpretation and usage.

A crucial concern in semantic modeling is the concept of semantic stability versus modulation. Cassani et al. (2023) delve into this issue using BERT-based embeddings, assessing the degree to which meanings fluctuate across different contexts. Their research shows that while certain words remain semantically stable across genres and registers, others—especially abstract or culturally sensitive terms—exhibit high modulation. This reinforces the need for fine-grained, context-aware modeling techniques in computational semantics, as traditional models that assign fixed meanings may miss crucial semantic variation inherent to natural language.

Psycholinguistic perspectives also contribute significantly to semantic studies. Aronowitz (2023) examines the boundary between episodic and semantic memory systems, arguing that semantic shift may occur not only in language but in the underlying memory structures that support language comprehension. This perspective blurs the line between linguistic meaning and cognitive representation, proposing that language change reflects deeper neurocognitive processes. Such an approach calls for interdisciplinary research linking linguistic theory,

neuroscience, and cognitive psychology to fully understand how semantic structures evolve over time.

The philosophical and theoretical discussion around "meaning" is enriched by Koslow (2022), who reflects on the nature of meaning change from a metaphysical standpoint. He challenges the assumption that meaning is a property that words "have" and instead presents meaning as an emergent relation involving speakers, contexts, and interpretative communities. His approach is semiotic and constructivist, providing a theoretical foundation for interpreting semantic shift not as a deviation but as the normative process of signification. This aligns with sociocultural approaches to semantics, where meaning is negotiated and redefined through discourse and social interaction.

In applied contexts, semantic models are increasingly being evaluated for robustness and reliability. De Jorge et al. (2023) question the semantic reliability of AI models in tasks such as image recognition and natural language understanding. They argue that semantic segmentation—assigning semantic labels to visual or linguistic input—is often flawed due to inconsistent or unstable meaning assignments in training data. This concern highlights a growing challenge: as AI systems become more semantically competent, their semantic models must be aligned with human interpretative expectations to ensure safe and meaningful interaction.

Concerns about semantic drift in large language models (LLMs) are further addressed by McIntosh et al. (2024), who warn against radical meaning shifts induced by reinforcement learning algorithms. Their research reveals that models trained with human feedback often amplify semantic biases and undergo untraceable semantic modulations, which can lead to hallucinations or misinterpretations in language generation tasks. This insight underscores the urgent need for more transparent and controlled semantic modeling in AI, particularly in high-stakes applications such as legal or medical domains.

International and multilingual perspectives also illuminate the study of semantic change. Costley, Kula, and Marten (2023) examine translanguaging practices in Zambia, revealing how public signage and multilingual communication contribute to novel semantic formations. In multilingual societies, words often carry layered or hybrid meanings that resist categorization within a single linguistic system. These findings challenge Eurocentric models of semantic analysis and advocate for a pluralistic, situated understanding of meaning that reflects global linguistic realities.

Finally, Kutuzov and Pivovarova (2021) provide a concrete example of cross-linguistic semantic shift detection in their RuShiftEval project, which focuses on semantic drift in Russian. Their study combines diachronic corpora and embedding models to detect changes in meaning across different time periods. Their methodology highlights the applicability of semantic shift detection tools beyond English and suggests that robust semantic theories must be validated in diverse linguistic contexts. Their work offers a blueprint for future research on semantic shift in under-resourced and morphologically rich languages.

In conclusion, the literature on semantic shift and modern semantics is both rich and varied, encompassing theoretical, historical, computational, pedagogical, and philosophical dimensions. It reveals a consensus that meaning is dynamic, context-sensitive, and socially constructed. Whether through historical borrowing, computational modeling, cognitive analysis, or sociolinguistic observation, the studies reviewed here collectively contribute to a deeper, more nuanced understanding of how language evolves over time. The integration of multiple methodologies—from corpus analysis to neural embeddings—has transformed semantic research into a vibrant interdisciplinary field capable of tracing meaning shifts with unprecedented precision and insight.

METHODS

This study adopts a qualitative library research method as its core approach to investigating modern semantics in relation to the shifting of language meaning. As a non-empirical method, library research relies entirely on the systematic collection, critical review, and interpretative synthesis of existing scholarly works, theories, and publications relevant to the research topic. The decision to use this methodology is grounded in the nature of semantic analysis itself, which requires extensive theoretical engagement, historical contextualization, and conceptual mapping, rather than field experimentation or data collection from human participants. This method is especially suitable for tracing semantic shift, as it allows the researcher to draw connections between past and contemporary theories, identify trends and transformations in linguistic thinking, and explore a wide range of conceptual and analytical frameworks presented by experts in the field of semantics and language change.

The process of this research began with the identification of relevant and authoritative sources, including peer-reviewed journal articles, academic books, conference proceedings, and preprint publications. These sources were selected based on their relevance to key themes such as semantic theory, lexical change, word embedding models, historical linguistics, pragmatics,

and computational approaches to meaning detection. Notably, this study integrates perspectives from well-established semantic theorists and contemporary scholars who examine semantic shift through both traditional linguistic theory and computational modeling. The materials reviewed span across multiple decades and disciplines, ensuring a comprehensive and multidimensional understanding of the topic. Primary criteria for source selection included scholarly credibility, recency (particularly in studies dealing with computational semantics), and thematic alignment with the research objective of analyzing how semantic shifts are studied and understood in modern semantics.

The analytical framework employed in this library research is primarily descriptive and interpretative. Rather than collecting new data or conducting statistical analyses, this study focuses on understanding, comparing, and interpreting the arguments and findings of prior research. The methodology involves carefully categorizing the literature according to thematic clusters—such as historical semantic change, cognitive semantics, corpus linguistics, semantic shift in large language models, and sociolinguistic influences on meaning. Within each category, the researcher analyzes how meaning shifts have been conceptualized, modeled, or observed by various authors. This technique allows for a layered and holistic understanding of how semantics has evolved as a discipline and how the phenomenon of meaning change is addressed in various contexts.

Moreover, the study makes use of conceptual synthesis, in which multiple theories and viewpoints are critically juxtaposed to highlight points of convergence, divergence, and theoretical evolution. For instance, classic semantic frameworks that emphasize referential meaning and fixed definitions are compared with more modern approaches that stress usage-based meaning, contextual interpretation, and computational flexibility. This methodological approach facilitates a diachronic analysis of semantic theory, tracing how the conceptualization of language meaning has shifted across time, influenced by broader linguistic, technological, and societal developments.

In addition, interdisciplinary integration forms a crucial aspect of the methodology. Since semantics is inherently connected to disciplines such as philosophy, psychology, artificial intelligence, and sociolinguistics, this study adopts an integrative stance that values cross-disciplinary insights. Works from computational linguistics, such as those exploring word embeddings or semantic drift in neural networks, are interpreted alongside theoretical linguistic discussions and historical analyses. This multidimensional reading strategy enables the

researcher to appreciate not only the depth of individual studies but also the broader trajectory of semantic thought and its applications in real-world language dynamics.

Finally, the methodology includes critical reflection on the strengths and limitations of the literature reviewed. The researcher not only synthesizes findings but also evaluates the assumptions, methodologies, and implications presented by each study. By identifying gaps in the literature and contrasting conflicting perspectives, the study aims to provide a well-rounded and critically informed analysis of how semantic shift is currently conceptualized and explored in modern semantics.

Subheading

FINDINGS AND DISCUSSION

1. Lexical Trajectories: Borrowings and Neologisms

The investigation into the evolution of English vocabulary reveals that semantic shift often follows pathways of borrowing and neologism, which serve as primary mechanisms for linguistic innovation. Mishra (2023) meticulously documents how borrowed words from other languages such as Latin, French, German, and more recently Arabic and Hindi, undergo meaning shifts once embedded into English grammatical and cultural contexts. For instance, "algorithm" retains its core mathematical meaning post-borrowing yet has contracted semantically to evoke ideas of mysterious digital processes or manipulative social media curation. This dual quality highlights that borrowed terms can simultaneously preserve original denotations while acquiring cultural connotations through generalized usage. Neologisms—especially technological ones like “selfie,” “cloud,” or “blockchain”—exhibit semantic expansion whereby their initial precise technical referents broaden to evoke entire experiential or aspirational domains tied to identity and control. Mishra’s findings indicate that the lifecycle of borrowings often transforms through stages of semantic shift, and that neologisms bypass historical nostalgia altogether, being born fully embedded in globalization discourses of innovation (Mishra, 2023).

Analysis of borrowings in Middle English, as explored by Sylvester, Tiddeman, and Ingham (2022), complements Mishra’s work by emphasizing domain specificity. Their domain analysis demonstrates that religious and legal terms borrowed post-Norman Conquest did not merely transplant from French to English; they were recontextualized within English semantic fields. For example, "justice" shifted from a generalized moral sense to a technical term within English legal discourse. The semantic narrowing here reflects the socio-functional restructuring of

power. Meanwhile, in non-technical domains such as hospitality or cooking, borrowings retained broader, more informally construed meanings. These findings illustrate that the fate of borrowings depends heavily on their pragmatic embedding and socio-institutional adoption (Sylvester et al., 2022).

The combination of Mishra's and Sylvester et al.'s insights underscores that semantic shift is rarely random—it's mediated by social domains, power structures, and the communicative needs of speakers. Borrowings become lexical carriers of ideological frames, and neologisms inject new categories into the cognitive-linguistic system. This dual process positions English not only as a linguistic sponge but also as a field where meanings are socially sculpted through contact, power, and praxis.

2. Cognitive Foundations of Semantic Change

To trace how semantic meaning shifts cognitively, Feist (2022) advances a model in which lexical items are cognitively anchored in prototypical structures, with meaning determined by dynamic relational mappings rather than static definitions. Feist explains that category membership is gradient, contingent on how well a word's semantic features align with cognitive prototypes. Semantic shifts occur when repeated usage contexts realign those cognitive features—e.g., when "virus" extends its prototype from virology to computer network threats, relying on associative similarity and metaphorical resonance, yet still tracking a core idea of "hidden, harmful agent" (Feist, 2022). Feist's model affords a principled account of semantic drift: shift is driven cognitively by distributed usage that reshapes prototype boundaries, reconciling theoretical fluency with empirical complexity.

Aronowitz (2023) complements this cognitive perspective by proposing that semanticization—where episodic memory content becomes integrated into semantic memory—can serve as a mechanism for meaning change. He argues that repeated social discourse can elevate episodic associations (e.g., "9/11" as an event) into semantic tokens carrying cultural meanings ("ground zero," "resilience," "terror"). This process destabilizes the monolithic boundary between episodic references and semantic base terms. Thus in the cognitive architecture, repeated contextual usage migrates experiential memory traces into the semantic system. This framework enhances our understanding of semantic shift by implicating memory systems as drivers of meaning change—where meaning shifts are not only lexical but also psychological (Aronowitz, 2023).

Koslow (2022) further addresses theoretical underpinnings of meaning change by framing meaning as emergent, not possessed. His semiotic-constructionist perspective contends that meaning arises from the interplay between speakers, contexts, and interpretative communities. Accordingly, semantic shifts reflect evolving social perspectives: words once neutral can adopt evaluative connotations (amelioration or pejoration) when used within changing interpretative frameworks. For example, “queer” shifted from derogatory to a derided term reclaimed as identity. Koslow illustrates how such shifts cannot be fully explained by etymology or frequency—they require an analysis of discursive power and social meaning-making.

Taken together, the works of Feist, Aronowitz, and Koslow construct a multi-layered cognitive discourse: words inhabit mental prototypes but are simultaneously shaped by memory systems and social negotiation. Semantic change thus occupies the intersection of cognitive realignment, episodic reclassification, and culturally mediated reinterpretation.

3. Interactional Semantics and Pragmatic Shift

Wedgwood’s (2021) interactional framework proposes that meaning emerges through usage, implicating conversation and inference as central mechanisms in semantic change. Rather than meaning being encoded and fixed in a lexicon, it is dynamically shaped by conversational repertoires, cooperative principles, and inference-driven reinterpretation. Wedgwood explores how indirect speech acts or implicatures can suddenly alter a word’s usage: for instance, “dropping someone” previously referred simply to refusing to meet someone, but in contemporary usage has shifted to a connotation of social exclusion within digital communities. This kind of meaning shift cannot be tracked via static definitions but only through patterns of implied use and conversational inference.

Guo, Xypolopoulos, and Vazirgiannis’s (2021) empirical study of COVID-19 discourse on Twitter offers a concrete demonstration of interactional semantic shift. By training contextual embedding models on Twitter timelines before and during the pandemic, they quantified semantic shifts in real-time. Words such as “case,” “positive,” and “mask” acquired pandemic-specific contextual vectors, reflecting social changes in how people used them. This pragmatically driven shift, prompted by collective interaction around a global event, crystallizes Wedgwood’s claims: meaning emerges via communal negotiation and inference under intense usage conditions (Guo et al., 2021).

These interactional and pragmatic models illustrate a compelling analytical trajectory: not only do actual utterance contexts matter—they can reshuffle semantic usage overnight under intense

communicative pressure. In such cases, meaning becomes contingent on collective experiential frames rather than stable definitions.

4. Computational Detection and Distributional Semantics

Montanelli and Periti (2023) provide a comprehensive overview of computational models for detecting semantic shift, underpinned by distributional semantics and embedding alignment across temporal slices. They show that model-driven detection of semantic movement can reveal both gradual drift and sharp meaning pivots, as language responds to shared experiences, fads, or sociocultural juxtapositions. Their survey of techniques—from dynamic alignment of vectors to density-based cluster tracking—reveals that embeddings like BERT, when compared over corpora sampled at different time intervals, map semantic trajectories that mirror both subtle re-assignment (e.g., "tweet" evolving from bird sound to short text post) and dramatic shifting (e.g., "Zoom" from verb to corporate noun). This computational capacity enables real-time tracing of meaning shifts at unprecedented scale (Montanelli & Periti, 2023).

Geeraerts et al. (2024) corroborate these findings, combining distributional models with theoretical semantic insights. Their investigation suggests that many lexical variations reflect changes in frequency and usage patterns rather than binary meaning flips. By analyzing corpora over time, they point out how certain words may develop peripheral senses that compete for dominance. The authors differentiate between stable core meaning and looser peripheral meaning, suggesting that semantic shift can be interpreted as reconfiguration within the semantic field rather than complete redefinition. This allows for a gradient understanding of semantic shift: instead of wholesale spectrum change, we observe shifts in weighting and salience within word meaning networks (Geeraerts et al., 2024).

Similarly, Cassani et al. (2023) examine semantic modulation and stability in BERT embeddings, targeting how the same word can express different nuances across registers or time periods. Their psycholinguistic approach emphasizes measuring vector dispersion across usage environments. They find that words with abstract or ideological content (e.g., "justice," "equality," "progress") show more dispersion than concrete nouns. From a computational standpoint, such dispersion signals semantic instability—an ongoing shift rather than endpoint change. This underscores the complexity of modeling semantic change in language models and reinforces the need for context-sensitive trajectory tracking (Cassani et al., 2023).

The RuShiftEval shared task (Kutuzov & Pivovarova, 2021) provides methodological validation for semantic shift detection in Russian. Their evaluation confirms that high-quality

diachronic embeddings can identify semantic change even in morphologically rich and syntactically flexible languages. Models successfully detect subtle shifts in meaning and rank them by change magnitude, demonstrating that computational techniques are transferable across linguistic boundaries. The ability of distributional models to trace semantic trajectories in under-resourced languages supports the universal applicability of modern computational semantics (Kutuzov & Pivovarova, 2021).

Taken together, the computational and distributional literature indicates that embedding-based methods are now central to tracing semantic shifts across contexts and time. These techniques reveal patterns of semantic movement, from core drift to peripheral expansion, allowing interdisciplinary alignment between theory, cognition, and computational modeling.

5. Educational and Epistemological Implications

Steingartner and Sivý (2024) examine how semantic change discoveries can be translated into pedagogical models. They argue that meaning shift analysis improves semantic insight if learners engage with dynamic semantic data through simulations or interactive exploration of corpora. Their educational interventions showed that exposing students to corpus-based semantic trajectories strengthened their meta-linguistic awareness and comprehension of meaning variability. They contend that involving learners in semantic investigation mirrors the research process, fostering deeper epistemological understanding of how meaning evolves in real time (Steingartner & Sivý, 2024).

Reilly et al. (2025) provide a meta-discursive contribution by offering a multidisciplinary glossary of semantic terminology. Their work has helped clarify the increasingly diverse semantic discourse by defining terms like “semantic drift,” “semantic bleaching,” “contextual modulation,” and “pragmatic denotation.” This terminological consolidation aids education and cross-disciplinary research by establishing a shared vocabulary. For scholars, this clarity reduces conceptual ambiguity and promotes coherent methodological integration, especially important when tracing meaning across multiple theoretical and empirical approaches (Reilly et al., 2025).

6. Semantic Reliability in AI Systems

De Jorge et al. (2023) identify a critical practical concern: flawed semantic segmentation in AI systems may result from unstable semantic representations. Their analysis shows that computer vision models trained on skewed or inconsistent semantic labels may exhibit classification

errors due to semantic ambiguity. Though their context is visual data, the principle transfers to NLP: unstable semantic trajectories in embeddings can lead to miscommunication or semantic hallucination. Their findings underscore the importance of tracking and constraining semantic drift in AI systems to align model semantics with human semantic stability (De Jorge et al., 2023).

McIntosh et al. (2024) reinforce this view by diagnosing semantic vulnerability in LLMs (large language models). They demonstrate that reinforcement learning from human feedback (RLHF) may produce unexamined meaning shifts in model behavior. Words like “violence,” “terrorism,” or “gender” may shift meaning subtly across prompts and regulatory frameworks, producing biased or unpredictable model outputs. Their critique highlights that meaning is not fixed in neural representations, and that unmonitored model updates can introduce semantic distortions. They urge for semantic auditing as a necessary aspect of responsible AI deployment (McIntosh et al., 2024).

Collectively, these works show that computational tracing of semantics not only informs theoretical linguistics but also demands accountability in AI systems where meaning matters. Ensuring semantic reliability requires modeling grammatical stability, detecting shift, and anchoring semantics in human interpretative norms.

7. Sociolinguistic and Multilingual Dimensions

Costley, Kula, and Marten (2023) offer an ethnographic perspective by demonstrating how translanguaging spaces—public writing, signage, advertising in Zambia—serve as fertile ground for semantic innovation. Words borrowed and hybridized in multilingual contexts reveal new semantic relationships that defy monolingual categorizations. Their mapping of Ndola’s urban signage demonstrates that such lexical fields attend simultaneously to English, Bemba, Kaonde, and Nyanja influences. Words like “shukani” (rise up) are used in local campaigns and gain added cultural meaning in their urban display. This shows that multilingual semantic innovation is not mere contact-induced borrowing but transformative re-anchoring of meaning within local practices (Costley et al., 2023).

Finally, Reilly et al.’s (2025) glossary advocates for cross-contextual sensitivity in semantic definitions, emphasizing that meaning is culturally contingent. Their inclusion of diverse linguistic case studies—from high-resource to under-resourced languages—underscores the importance of contextual semantics and warns against imposing one-size-fits-all definitions.

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This reflects a growing trend in semantics research to respect diversity in usage, ideology, and cultural meaning construction.

CONCLUSION

Language meaning is constantly evolving, influenced by various social, cultural, and technological factors. Modern semantic studies have revealed that word meanings are not fixed; instead, they adapt over time to meet the changing communicative needs of speakers. This shift in meaning demonstrates that language is a living, dynamic system shaped by ongoing human interaction and interpretation.

One of the main drivers of semantic shift is the introduction of loanwords and the creation of new terms. When foreign words are adopted into a language, their meanings often shift to align with the receiving culture's context. Similarly, neologisms—newly coined words—often emerge in response to technological innovations or cultural phenomena, and their meanings may quickly expand, narrow, or transform based on how they are used in society.

Cognitive factors also play a crucial role in meaning change. As human understanding of concepts evolves, so does the way people use language to express those concepts. Frequent use of certain words in specific contexts can lead to a collective reinterpretation of their meanings. This highlights the deep connection between language, thought processes, and experiential learning in semantic development.

Social interaction and pragmatic usage are equally central in influencing semantic change. In everyday communication, meanings can shift due to sarcasm, metaphor, or contextual reinterpretation. Thus, the meaning of a word is not solely defined by its dictionary entry, but also by the speaker's intention and the listener's interpretation within a given situation.

Technological advances, especially the rise of social media and digital communication, have dramatically accelerated the pace of semantic shift. Words can now change meaning rapidly as they are disseminated across digital platforms. Computational tools like language models and semantic embedding systems allow researchers to trace these shifts in real-time, offering valuable insights into how meanings evolve in digital discourse.

The implications of semantic shift extend far beyond linguistics. Understanding how meanings change is vital in education, media literacy, and artificial intelligence. In AI development, for example, tracking semantic nuances ensures that systems better interpret user intentions and cultural contexts. Thus, modern semantic analysis has practical applications that affect diverse sectors of society.

In conclusion, modern semantics underscores that word meaning is inseparable from context. As societies evolve, so too does language, mirroring shifts in thought, interaction, and culture. Exploring meaning change requires an interdisciplinary approach, drawing from linguistics, cognitive science, technology, and sociology to fully grasp how and why language transforms. This comprehensive understanding is essential to keep pace with the living nature of human communication.

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